

Erdős Problem #1040

EP-1040

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $F \subseteq \mathbb{C}$ be a closed infinite set, and let $\mu(F)$ be the infimum of $|\{z : |f(z)| < 1\}|$, as f ranges over all polynomials of the shape $\prod (z - z_i)$ with $z_i \in F$.

Is $\mu(F)$ determined by the transfinite diameter of F ? In particular, is $\mu(F) = 0$ whenever the transfinite diameter of F is ≥ 1 ?